

8. Cyclomatic Complexity in Software Testing is a testing metric used for measuring the complexity of a software program. It is a quantitative measure of independent paths in the source code of a software program. Cyclomatic complexity can be calculated by using control flow graphs or with respect to functions, modules, methods or classes within a software program. Cyclomatic complexity for a flow graph G is $V(G)=E-N+2$. Cyclomatic complexity is a software metric (measurement), used to indicate the complexity of a program, $\text{Cyclomatic complexity} = E - N + P$ where, E = number of edges in the flow graph, N = number of nodes in the flow graph, P = number of nodes that have exit points. Find Cyclomatic Complexity for a graph having number of edges as 17, number of nodes as 13 and number of predicate nodes in the flow graph as 5.

Algorithm: Cyclomatic Complexity

1. Start.
2. Input the number of edges **Ed**.
3. Input the number of nodes **Node**.
4. Input the number of predicate nodes **P**.
5. Calculate Cyclomatic Complexity using:
 - **$CC = E - N + 2$**
 - *(or verify using **$CC = P + 1$**)*
6. Display **Cyclomatic Complexity = CC**.
7. Stop.

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main

```
graph TD; start([start]) --> GET_node[/GET node/]; GET_node --> GET_edge[/GET edge/]; GET_edge --> GET_pr[/GET pr/]; GET_pr --> cc_assignment[cc = edge-node+2]; cc_assignment --> cc_update[cc = cc+pr+2]; cc_update --> PUT_ccf[/PUT ccf/]; PUT_ccf --> end([end]);
```

node: 5
edge: 6
pr: 3
cc: 8

Console

```
7  
--- Run Complete! 8 Symbols  
Evaluated ---  
8  
--- Run Complete! 8 Symbols  
Evaluated ---
```

Presentation ready, set, go

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